

Day 02 – Embeddings & Data Source Collection for SLM

Name: Moturi Lalitha Sowjanya

Team: 4

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1) OBJECTIVE

To understand the concept of embeddings, their role in language models, how they differ from tokenizers, and their real-world applications. Also, to identify suitable data sources for the proposed SLM and document how they contribute to its development.

2) RESEARCH TOPICS

2.1. Embeddings

2.1.1. What are embeddings?

- Embeddings are numerical representations (vectors) of words, sentences, or documents that capture their meaning and relationships. They convert text into numbers so that language models can understand and compare the meaning of different pieces of text.
 - Words or sentences with similar meanings have embeddings that are close to each other in the vector space.
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2.1.2. Why are they needed in Language Models?

Embeddings are needed because language models cannot understand text directly. They work with numbers, so embeddings convert text into numerical vectors while preserving its meaning.

They help language models:

- Understand the meaning of words and sentences.
 - Identify relationships between similar words.
 - Compare the similarity between different texts.
 - Perform tasks such as semantic search, question answering, and text classification.
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2.1.3 How do Embeddings Differ from Tokenizers?

Tokenizer	Embedding
A tokenizer splits text into smaller units called tokens.	An embedding converts tokens into numerical vectors.
It is the first step in processing text.	It comes after tokenization.
It prepares the text for the language model.	It helps the model understand the meaning and relationships between words.
Output: Tokens (words, sub words, or characters).	Output: Numerical vectors (embeddings).

Example:

Input Text:

"I love AI"

Tokenizer Output:

["I", "love", "AI"]

Embedding Output:

"I" → [0.12, -0.45, 0.81, ...]

"love" → [0.67, 0.15, -0.28, ...]

"AI" → [0.94, -0.10, 0.52, ...]

In simple words:

- **Tokenizer** breaks the text into smaller pieces.
- **Embedding** converts those pieces into numbers while preserving their meaning, so the language model can understand them.

2.1.4. Real-world applications of embeddings

1. **Semantic Search** – Finds information based on meaning rather than exact keywords.
2. **Question Answering Systems** – Retrieves relevant information to answer user questions.
3. **Chatbots and Virtual Assistants** – Helps understand user queries and generate accurate responses.
4. **Recommendation Systems** – Suggests similar products, movies, books, or songs based on user preferences.
5. **Document Similarity** – Compares documents to identify similar content.

6. **Machine Translation** – Helps translate text from one language to another while preserving meaning.
7. **Retrieval-Augmented Generation (RAG)** – Converts documents into embeddings so the system can retrieve the most relevant information before generating a response.

3) DATA COLLECTION FOR THE PROPOSED SLM

3.1. Collected Data Sources & Why These Data Sources are Useful

Data Source	Type	Description	Why is it Useful for the Proposed SLM?
WHO Health Topics (Fact Sheets)	Website	Contains information on diseases, symptoms, diagnosis, treatment, prevention, public health guidelines, and health awareness.	Provides trusted information about diseases, symptoms, treatments, and prevention, helping the SLM answer healthcare questions accurately.
MedlinePlus	Website	Contains health topics, drugs & supplements, genetics, medical tests, and a medical encyclopedia.	Provides easy-to-understand information about diseases, medicines, medical tests, and health topics, helping the SLM give simple and clear answers.
CDC (Centers for Disease Control and Prevention)	Website	Contains information on diseases and conditions, healthy living, vaccines, environmental health, workplace safety, and global health.	Provides updated information on diseases, vaccines, and healthy living, helping the SLM provide reliable healthcare information.
Medical Question Answering Dataset	Datasets	Contains medical questions and answers from multiple healthcare domains, including cancer, diabetes, heart diseases, neurological disorders, disease prevention, genetics, and senior health.	Helps the proposed Medical Information Assistant SLM learn to understand and answer healthcare-related questions across different medical topics.

Data Source	Type	Description	Why is it Useful for the Proposed SLM?
Symptom Disease Dataset		Contains patient symptom descriptions along with their corresponding disease labels. The dataset helps identify diseases based on symptoms.	Helps the proposed Medical Information Assistant SLM understand the relationship between symptoms and diseases, allowing it to provide more accurate healthcare-related responses.

Links:

Websites:

- <https://www.who.int/news-room/fact-sheets>
- <https://medlineplus.gov/>
- <https://www.cdc.gov/health-topics.html>
- <https://www.cdc.gov/vaccines/index.html>
- <https://www.cdc.gov/vaccines/site.html#hcp>

Datasets:

- <https://www.kaggle.com/datasets/gvaldenebro/cancer-q-and-a-dataset>
- <https://huggingface.co/datasets/dux-tecblic/symptom-disease-dataset/viewer>